Separated voices and speaker timelines

Simulated test conversation: real Hindi speech, two speakers, heavy overlap and 5 dB market noise.

M1 Separation / Vismay

1 Mixed recording



26.4–44.8 s
Original mixture

Click to play

2 Speaker A

▶ Speaker A — separated voice

3 Speaker B

▶ Speaker B — separated voice

+9.6 dB SI-SDR improvement on 245 test overlap regions

M2 Diarization / Darshan

Actual timeline.json output (excerpt)

Start (s)	End (s)	Speaker
26.85	31.75	Speaker_A
31.75	33.20	Speaker_B
33.20	40.20	Speaker_A

20.3% DER

Spectral clustering, 72 test conversations
GMM: 20.3% / pyannote 3.1: 18.4%

What the system actually transcribes

M3 / Abhinav IndicConformer speech recognition, Hinglish text and subtitle files

SAVED HINGLISH TRANSCRIPT

00:27–00:32 Speaker_A

sir mere koi galti hui mere haath se wo mera aai D card gum ho gaya hai sir

00:32–00:33 Speaker_B

haan aur baki sun

00:51–00:54 Speaker_A

to ab kab tak kitne baje tak office khula hai sir apna

Devanagari + Hinglish + SRT + JSON

21.0% word error

IndicWav2Vec: 38.0%
400 Vaani clips, single-speaker ASR

FULL PIPELINE / 72 CONVERSATIONS

Order	cpWER
A: separate first	55.8%
B: diarize first	49.5%

B wins in 50 of 72 conversations.
cpWER measures words and speaker attribution. Lower is better.

The report output, and what remains

M4 / Shrivaths Airavata 7B, quantised to 4 bits and run locally

ACTUAL REPORT EXCERPTS

Key point

“Speaker A lost his ID card”

Line 13 / Speaker_A / 00:51–00:54

“The office is open until what time?”

Produces a summary, keywords and translated dialogue.
Speakers and times pass through unchanged.

Measured limits

Transcript repair adds 0.9 cpWER points
(paired mean across 63 conversations).
No accepted action items in 252 reports.

Next steps

Improve Hindi speaker embeddings.
Evaluate a larger LLM for action items.
Retrain separation on Indian voices and noise.

These are planned extensions, not measured results.